

YOUNGJAE MIN (민영제)

📍 51 Vassar Street, 45-541K, Cambridge, MA 02139, USA

✉ yjm@mit.edu | 🌐 youngjae-min.github.io | 🐙 github.com/youngjae-min

RESEARCH INTERESTS	Reliable AI & Autonomy, Learning-Based Control, Continual Learning, Optimization I develop learning-enabled control methods for reliable intelligent systems that are safe, adaptive, and deployable in safety-critical settings. My work integrates machine learning, control, and optimization to design principled algorithms with theoretical guarantees and practical reliability.
EDUCATION	Massachusetts Institute of Technology Ph.D. Candidate in Autonomous Systems and Statistics ◦ Major: Machine Learning / Minor: Control Massachusetts Institute of Technology S.M. in Aeronautics and Astronautics Korea Advanced Institute of Science and Technology B.S. in Electrical Engineering, Mathematical Sciences (<i>summa cum laude</i>) June 2023 - (Dec. 2026) Advisor: Professor Navid Azizan Sep. 2021 - May 2023 Mar. 2014 - Feb. 2020* (*two years in military)
RESEARCH EXPERIENCE	Laboratory for Information and Decision Systems, MIT <i>Research Assistant</i> / Advisor: Prof. Navid Azizan ◦ Learning and control with provable input-output constraint satisfaction [P1, P3, C7] ◦ Online/continual learning robust to catastrophic forgetting [P4, C6, C9, J5] CI Mobility Research Team, Honda Research Institute, CA <i>Research Intern</i> / Advisor: Sangjae Bae ◦ Motion planning via integrated offline RL and online planning [P2] Laboratory for Information and Control Systems, KAIST <i>Undergrad. Researcher</i> / Advisor: Prof. Han-Lim Choi ◦ Real-time 3-D mapping of dynamic environments [C8, C11] ◦ Learning and planning of dynamical systems [C12, J10, J14] Inference and Information for Data Science Lab, KAIST <i>Undergrad. Researcher</i> / Advisor: Prof. Hye Won Chung ◦ Representational capability of neural networks [C13] Signal Kinetics Group, Media Lab, MIT <i>Visiting Researcher</i> / Advisor: Prof. Fadel Adib ◦ Non-contact vital sign monitoring via mmWave radar Networking & Mobile Systems Lab, KAIST <i>Undergrad. Researcher</i> / Advisor: Prof. Sung-Ju Lee ◦ Indoor person localization via Wi-Fi signals Sep. 2021 - (Dec. 2026) May 2025 - Aug. 2025 Feb. 2019 - July 2021 Mar. 2018 - Jan. 2019 June 2018 - Sep. 2018 June 2017 - Feb. 2018
PUBLICATIONS	(<i>P: preprint, C: conference proceedings, J: journal articles, *equally contributed</i>) [P1] “HardNet: Hard-Constrained Neural Networks with Universal Approximation Guarantees,” <i>preprint</i> Youngjae Min , Navid Azizan [arXiv: 2410.10807] NeurIPS Constrained ML Workshop 2025 (selected as Oral)

- [P2] “HOLO-MPPI: Multi-Scenario Motion Planning via Hierarchical Policy Optimization,” *preprint*
Youngjae Min, Jovin D’sa, Faizan M Tariq, David Isele, Navid Azizan, Sangjae Bae
- [P3] “HardNet++: Nonlinear Constraint Enforcement in Neural Networks,” *preprint*
 Andrea Goertzen, Kaveh Alim, **Youngjae Min**, Navid Azizan [\[arXiv: 2604.19669\]](#)
- [P4] “ORFit: Efficient One-Pass Learning for Overparameterized Models,” *preprint*
Youngjae Min, Namhoon Cho, Navid Azizan
- [J5] “SketchOGD: Memory-Efficient Continual Learning”
Youngjae Min, Benjamin Wright, Jeremy Bernstein, Navid Azizan
TSP 2026 - IEEE Transactions on Signal Processing [\[arXiv: 2305.16424\]](#)
- [C6] “II-ORFit: One-Pass Learning with Bregman Projection”
 Namhoon Cho, **Youngjae Min**, Hyo-Sang Shin, Navid Azizan
ACC 2024 (Invited Session) - American Control Conference
- [C7] “Data-Driven Control with Inherent Lyapunov Stability”
Youngjae Min, Spencer M. Richards, Navid Azizan
CDC 2023 (Invited Session) - IEEE Conference on Decision and Control [\[arXiv: 2303.03157\]](#)
- [C8] “DS-K3DOM: 3-D Dynamic Occupancy Mapping with Kernel Inference and Dempster-Shafer Evidential Theory,” Juyeop Han*, **Youngjae Min***, Hyeok-Joo Chae, Byeong-Min Jeong, Han-Lim Choi
ICRA 2023 - IEEE International Conference on Robotics and Automation [\[arXiv: 2209.07764\]](#)
- [C9] “One-Pass Learning via Bridging Orthogonal Gradient Descent and Recursive Least-Squares”
Youngjae Min, Kwangjun Ahn, Navid Azizan
CDC 2022 (Invited Session) - IEEE Conference on Decision and Control [\[arXiv: 2207.13853\]](#)
- [J10] “Online Gaussian Process State-Space Model: Learning and Planning for Partially Observable Dynamical Systems,” Soon-Seo Park, Young-Jin Park, **Youngjae Min**, Han-Lim Choi
IJCAS 2022 - International Journal of Control, Automation and Systems [\[arXiv: 1903.08643\]](#)
- [C11] “Kernel-Based 3-D Dynamic Occupancy Mapping with Particle Tracking”
Youngjae Min, Do-Un Kim, Han-Lim Choi
ICRA 2021 - IEEE International Conference on Robotics and Automation
- [C12] “Informative Planning of Mobile Sensor Networks in GPS-Denied Environments”
Youngjae Min, Soon-Seo Park, Han-Lim Choi
SciTech 2020 - AIAA Science and Technology Forum and Exposition [\[arXiv: 1909.11046\]](#)
- [C13] “Shallow Neural Network can Perfectly Classify an Object following Separable Probability Distribution”
Youngjae Min, Hye Won Chung
ISIT 2019 - IEEE International Symposium on Information Theory [\[arXiv: 1904.09109\]](#)
- [J14] “A Distributed ADMM Approach to Non-Myopic Path Planning for Multi-Target Tracking”
 Soon-Seo Park*, **Youngjae Min***, Jung-Su Ha, Doo-Hyun Cho, Han-Lim Choi
Access 2019 - IEEE Access [\[arXiv: 1807.11068\]](#)

HONORS & AWARDS	MIT SoE MathWorks Fellowship, <i>MIT School of Engineering & MathWorks Inc.</i>	2025 - 2026
	KEF Scholarship, <i>Kwanjeong Educational Foundation</i>	2023 - 2027
	Global Leadership Award, <i>KAIST</i>	2020
	◦ Awarded to the top undergraduate for outstanding creativity	
	GE Foundation Scholar-Leaders Program, <i>Fulbright / GE Foundation</i>	2015 - 2019

	KFAS Undergraduate Student Scholarship, <i>Korea Foundation for Advanced Studies</i>	2015 - 2019
	KAIST Presidential Fellowship, <i>KAIST</i>	2014
	◦ Awarded to the top 11 students of the Class of 2018	
PROFESSIONAL ACTIVITIES	Reviewing : American Control Conference (ACC), IEEE Conference on Decision and Control (CDC), IEEE Control Systems Letters (L-CSS), IEEE International Conference on Robotics and Automation (ICRA), IEEE Robotics and Automation Letters (RA-L), IEEE Transactions on Robotics (T-RO), International Conference on Learning Representations (ICLR), Learning for Dynamics & Control (L4DC)	
TEACHING EXPERIENCE	2.004 Dynamics and Control II, Teaching Assistant, <i>MIT</i>	Fall'22
	PH141 General Physics I, Tutor, <i>KAIST</i>	Spring'15, Spring'17
	KAIST Global Institute for Talented Education, Online Tutor, <i>KAIST</i>	Aug. 2014 - July 2015
EXTRACURRI- CULARS	Charles River Running Crew, President, <i>MIT</i>	May 2024 - May 2025
	KAIST EE Newsletter, Reporter, <i>KAIST</i>	Mar. 2017 - Dec. 2017
	ROK Army & U.S. Army, IT Specialist (Sergeant, KATUSA), <i>Cp. Carroll</i>	July 2015 - Apr. 2017
	The Real LUNATIC, B-Boy, <i>KAIST</i>	Mar. 2014 - July 2015
SKILLS	Programming Python, Pytorch, C/C++, MATLAB, ROS, CUDA	